A Structural VAR Analysis of The Phillips Curve

University of Bologna

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Shahriyar Majidi, Ekaterina Nemchenko, Jelena Cesnokova

Abstract

This study analyzes the relationship between unemployment and inflation in the U.S. economy through the lens of the Phillips Curve, using a structural vector autoregressive (SVAR) model. Leveraging time-series data from the Federal Reserve Bank of St. Louis, this analysis spans multiple economic indicators, including the Federal Funds Rate, core PCE, unemployment rate, and real GDP. Following data pre-processing to ensure stationarity, a reduced-form VAR model is applied, and impulse-response functions are estimated to observe the temporal effects of shocks on unemployment. Results reveal a short-lived, negative relationship between inflation and unemployment, with impacts dissipating over 10 quarters. The study contributes to the ongoing evaluation of the Phillips Curve's relevance in macroeconomic policy.

Keywords: Phillips curve, VAR, Structural VAR, Time-series analysis, Impulse-Response function.

1 Introduction

Inflation and unemployment are among the most closely watched macroeconomic variables in all economies around the world. Given that they are both measurements of the overall state of the economy, it is reasonable to assume that they are somehow correlated. The first to examine the relationship between unemployment and inflation was Bill Phillips. In his paper "The Relation Between Unemployment and the Rate of Change of Money Wage Rates in the United Kingdom" in 1958, Phillips found that unemployment and the rate of change of wage rates are negatively correlated in the British economy. Samuelson and Solow then took a step further and linked the increasing wage rates to inflation, showing that the same negative relationship holds between unemployment and inflation. The mathematical model that characterizes the relationship between these two variables is known as the Phillips curve. These findings about the Phillips curve led economists to believe that unemployment can be persistently reduced at the cost of increasing price levels in the economy. In the late 1960s however, Friedman and Phelps made the assertion that this relationship holds only in the short run, and that inflation cannot continue to reduce unemployment in the long run. Subsequent empirical works confirmed this claim, and later on, in his celebrated treatise, the Lucas Critique, Robert Lucas argued that this relationship between unemployment and inflation would break if any attempt is made by the policy makers to exploit it. The reason for that is simple: once the public realizes that the increasing prices are due to an expansionary monetary policy and not increased economic activity, all expectations are adjusted, and decisions in the labor market are made accordingly. Nonetheless, the Phillips curve is still monitored by central banks to assess the impact of their monetary policies on the labor market in the short run.

In this paper, we study the temporal extent of the trade-off between unemployment and inflation in the United States Economy by the means of a structural vector autoregressive model defined over a panel of time series data compiled from the Federal Reserve Bank of St. Louis. The data set consists of quarterly observations of the following variables:

1. Effective Federal Funds Rate.
2. Seasonally adjusted core PCE.
3. Seasonally adjusted unemployment rate.
4. Seasonally adjusted real GDP in billions of chained 2017 US dollars.

2 Data Pre-Processing

In the original data set, observations are recorded since the first quarter of 1959 up to the present day. To carry out this analysis however, all observations prior to the structural break of the U.S. GDP growth rate in 1985Q1 and after the outbreak of the COVID-19 pandemic in 2019Q4 were dropped from the data set. Moreover, in order to achieve stationarity, the original variables were transformed according to table 1.[1] The log differences of the real GDP and PCE are simply equal to the GDP growth and inflation rates. It is important to point out that taking differences does not affect the interpretability of an autoregressive model. Taking differences results in the loss of

[1] We imployed the transformations recommended by (McCracken and NG, 2015.).

information about the levels of the variables, not the coefficients of the model. The plot

Original Variable	Symbol	Transformed Variable	Transformation
Federal Funds Rate	r_t	$\rightarrow i_t := \Delta r_t$	First Difference
Unemployment Rate	u_t	$\rightarrow w_t := \Delta u_t$	First Difference
Real GDP	Y_t	$\rightarrow y_t := \Delta ln(Y_t)$	Log Difference
Core PCE	Π_t	$\rightarrow \pi_t := \Delta^2 ln(\Pi_t)$	Second-order Log Difference

Table 1: Time Series Transformations

of all four time series after transformation are displayed in figure 1. In order to verify

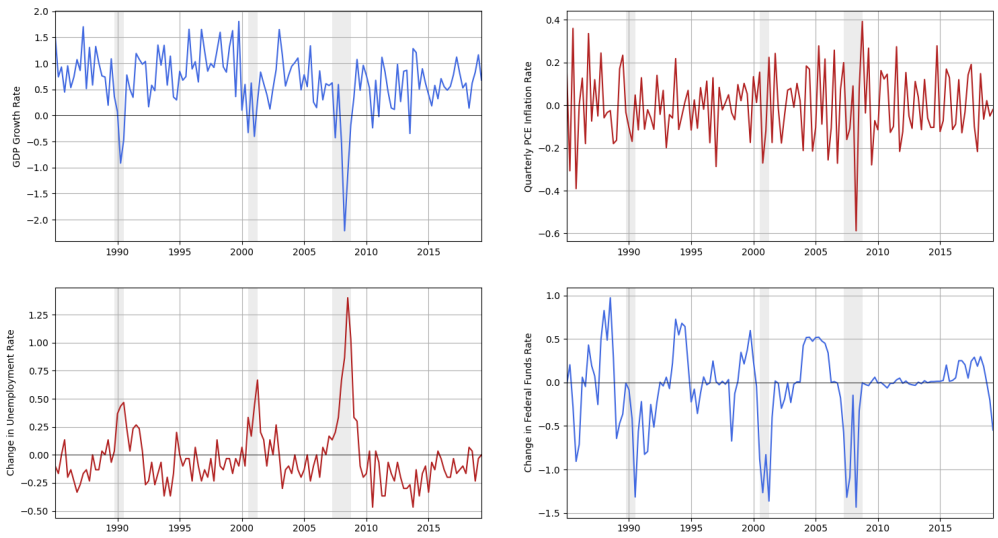


Figure 1: Time Series Data. Shaded regions are recession periods.

the absence of unit roots in the transformed series, we tested them with the Augmented Dickey-Fuller test The results are reported in table 2. In all cases, the null hypothesis is rejected at the 5% significance level. Therefore we regard the transformed variables as stationary.

	y_t	π_t	w_t	i_t
ADF test statistic	-5.07	-8.71	-3.45	-5.15
p-value	0.00	0.00	0.01	0.00
# lags used	1	4	8	0
# observations	132	129	125	133
critical value (5%)	-2.88	-2.88	-2.88	-2.88

Table 2: Augmented Dickey-Fuller test results.

3 Reduced Form Analysis

The Phillips curve is traditionally defined as:

$$w_t = \phi + \lambda \pi_t + \gamma^T X_t + \eta_t \quad \eta_t \sim \mathcal{N}(0,1) \tag{1}$$

Where w_t is the unemployment rate, ϕ the constant term, π_t the inflation rate, X_t a vector of control variables, and η_t a white noise. With the data set at hand, the vector of control variables includes the GDP growth rate and the Federal Funds rate. In the context of vector autoregressive models the Phillips curve can be modeled as:

$$Y_t = \Phi + \sum_{k=1}^{q} \boldsymbol{A}_k L^k Y_t + \varepsilon_t \quad \varepsilon_t \sim MDS(0, \boldsymbol{\Sigma_\varepsilon}) \tag{2}$$

Where Y_t is the vector of transformed variables and ε_t the vector of error terms[2]. The discussion on the ordering of the variables in the VAR is deferred to section 4. The lag order of the model is chosen to be 2 using the Akaike Information Criterion.[3] The OLS estimates of the constant vector and the matrix of coefficients are reported in table 3. Even though the first three variables seem to have no correlation at all, they do

Variable	Constant	L1.i_t	L1.π_t	L1.y_t	L1.w_t	L2.i_t	L2.π_t	L2.y_t	L2.w_t
i_t	-0.01	0.62	-21.46	9.85	-0.37	-0.01	-3.64	-11.23	0.15
	(0.81)	(0.00)	(0.27)	(0.12)	(0.04)	(0.91)	(0.85)	(0.08)	(0.35)
π_t	0.00	0.00	-0.49	0.29	0.00	0.00	-0.29	-0.03	0.00
	(0.85)	(0.12)	(0.00)	(0.66)	(0.04)	(0.41)	(0.00)	(0.36)	(0.02)
y_t	0.00	0.00	0.00	0.26	0.00	0.00	0.12	0.18	0.00
	(0.00)	(0.71)	(0.99)	(0.01)	(0.51)	(0.33)	(0.70)	(0.08)	(0.67)
w_t	0.07	-0.02	-22.46	-10.67	0.40	-0.04	-17.16	-2.87	0.13
	(0.03)	(0.76)	(0.04)	(0.00)	(0.00)	(0.45)	(0.12)	(0.42)	(0.15)

Table 3: OLS estimates of the VAR rounded up to two decimal points. P-values are in parentheses.

Granger-cause the change in unemployment rate and that is why they are included in the model. The partial autocorrelation function of the residuals of all four equations are shown in figure 2. The residuals of Federal Funds rate and GDP growth rate exhibit no serial autocorrelation, however there are signs of weak autocorrelation in the residuals of inflation and unemployment. To explore the issue of autocorrelation in the residuals in more depth, we constructed the null hypothesis: H_0: residuals do not exhibit serial correlation up to 12 lags, and tested the null using the Ljung-Box test. The results are reported in table 4. The null is not rejected at the 5% significance level, however, we acknowledge that residuals of inflation barely survive the Ljung-Box test.

[2]No prior distributions are assumed for the error terms, they are only required to be martingale difference sequences.

[3]Even though BIC is generally preferred over AIC in high dimensional settings, we used the AIC because first, this sample is far from high-dimensional, and second, the residuals of the model that the BIC chooses are serially correlated whereas the residuals of the model that the AIC chooses are not.

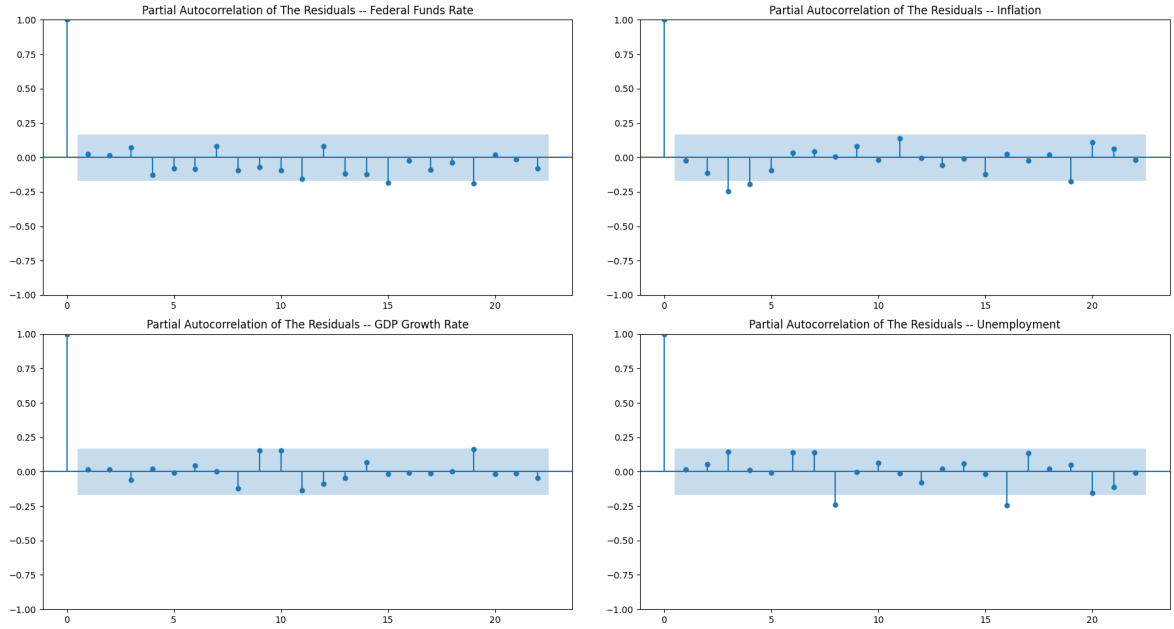


Figure 2: Partial autocorrelation of residuals.

	ε_t^i	ε_t^π	ε_t^y	ε_t^w
LB-Statistic	11.65	20.52	12.47	17.89
p-value	(0.47)	(0.06)	(0.41)	(0.12)

Table 4: Results of the Ljung-Box test.

4 Structural Analysis

In a structural VAR model, reduced form shocks are related to structural shocks through the Cholesky factor of the variance-covariance matrix of the reduced form errors given that the variables are ordered appropriately in increasing degree of endogeneity. Even though one can find many data-driven methods in the literature that claim to have found the optimal ordering of the variables, this ordering must have its foundations in the macroeconomic theory, and applying data-driven techniques to carry out this task is simply misguided for one can find any desired ordering given different prior assumptions. Moreover, the results of these data-driven methods are highly unstable and change drastically given the slightest change in assumption, which is further testimony to the failure of such methodologies. Therefore, in order to identify the optimal ordering of the variables, we rely on theoretical reasoning and use as our guide the results of macroeconomic theory.

The most exogenous variable in this model is the Federal funds rate for it is the only variable that is directly controlled by the Federal Reserve. The next most exogenous variable is the inflation rate, the reason for that being that first, inflation is heavily influenced by the amount of money in circulation which is under the direct control of the open market committee, and second, nominal rigidity makes this variable relatively unresponsive to other economic shocks such as output shocks and labor market shocks in the short run. As for the remaining variables, unemployment rate is the more endogenous because by definition it is a direct consequence of reduced economic activity which is manifest in the GDP growth rate. The assumptions made above imply the following

ordering of the variables in the VAR model:

$$Y_t \equiv \begin{pmatrix} i_t \\ \pi_t \\ y_t \\ w_t \end{pmatrix} \quad \longrightarrow \begin{pmatrix} \varepsilon_t^i \\ \varepsilon_t^\pi \\ \varepsilon_t^y \\ \varepsilon_t^w \end{pmatrix} = \begin{pmatrix} p_{11} & 0 & 0 & 0 \\ p_{21} & p_{22} & 0 & 0 \\ p_{31} & p_{32} & p_{33} & 0 \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix} \begin{pmatrix} u_t^i \\ u_t^\pi \\ u_t^y \\ u_t^w \end{pmatrix} \tag{3}$$

Where ε_t denotes reduced form shocks and u_t the structural shocks. The Cholesky factor is estimated by the Cholesky decomposition of the variance-covariance matrix of the residuals:

$$\hat{\boldsymbol{\Sigma}}_{\boldsymbol{\varepsilon}} = PP^T \Rightarrow P = \begin{pmatrix} 0.32 & 0 & 0 & 0 \\ 0.00 & 0.00 & 0 & 0 \\ 0.00 & 0.00 & 0.00 & 0 \\ -0.05 & -0.01 & -0.07 & 0 \end{pmatrix} \tag{4}$$

This result implies that a structural shock on each of the first three variables has a negative impact on unemployment, implying a trade-off in the short run. The VMA representation of the model can be constructed by first rewriting the VAR in the companion form:

$$\begin{pmatrix} Y_t \\ Y_{t-1} \end{pmatrix} = \begin{pmatrix} \boldsymbol{A_1} & \boldsymbol{A_2} \\ \boldsymbol{I} & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} Y_{t-1} \\ Y_{t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ 0 \end{pmatrix} \tag{5}$$

The VMA representation of the VAR in the companion form is:

$$\boldsymbol{Y_t^*} = \sum_{k=0}^{\infty} \boldsymbol{J}^k L^k \boldsymbol{\varepsilon_t^*} \tag{6}$$

Where $\boldsymbol{J}$ is the matrix of coefficients multiplying the lagged regressors in 5. By defining the selection matrix as $R_1^T = \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \end{pmatrix}$ The VMA representation of the original VAR model is:

$$Y_t = \sum_{k=0}^{\infty} R_1^T \boldsymbol{J^k} R_1 L^k \varepsilon_t = \sum_{k=0}^{\infty} \left(R_1^T \boldsymbol{J^k} R_1 P \right) L^k u_t \tag{7}$$

Where $\left(R_1^T \boldsymbol{J^k} R_1 P \right) \equiv C_k$ is the matrix of dynamic multipliers. The ij^{th} element of this matrix gives the response of variable i to a structural shock on variable j, k quarters ahead.

$$C_k^{ij} = \frac{\partial}{\partial u_t^j} \mathbb{E}\left[Y_{t+k}^i | I_t \right] \tag{8}$$

The dynamic multipliers of unemployment up to four quarters ahead are reported in table 5.

	0	1	2	3	4
u_t^i	-0.048	-0.139	-0.127	-0.106	-0.076
u_t^π	-0.01	-0.004	-0.002	-0.001	-0.001
u_t^y	-0.069	-0.027	-0.015	-0.007	-0.005

Table 5: Dynamic multipliers of unemployment.

5 Impulse Response Analysis

The impulse response functions are estimated using the VMA representation of the VAR in 7. They forecast the response of a variable of interest to a unit shock in each of the variables. The estimated impulse responses and the 95% confidence bands of all variables are displayed in figure 3. Each row contains the graph of the estimated impulse response of one variable to a unit shock in all the other variables. Plots on the diagonal are the impulse responses of each variable to a shock in itself. As can be seen, all shocks are completely absorbed after about 12 quarters as expected. The negative relationship between unemployment and the other variables is evident in the last row of the figure. The impulse response of unemployment to inflation depicts the temporal extent of the trade-off between the two variables. It is apparent from this figure that the negative relationship between inflation and unemployment is only short lived, and after about 10 quarters unemployment returns to its natural rate. Even though this negative relationship does exist, it is very small in magnitude, and after the second quarter it becomes statistically insignificant.

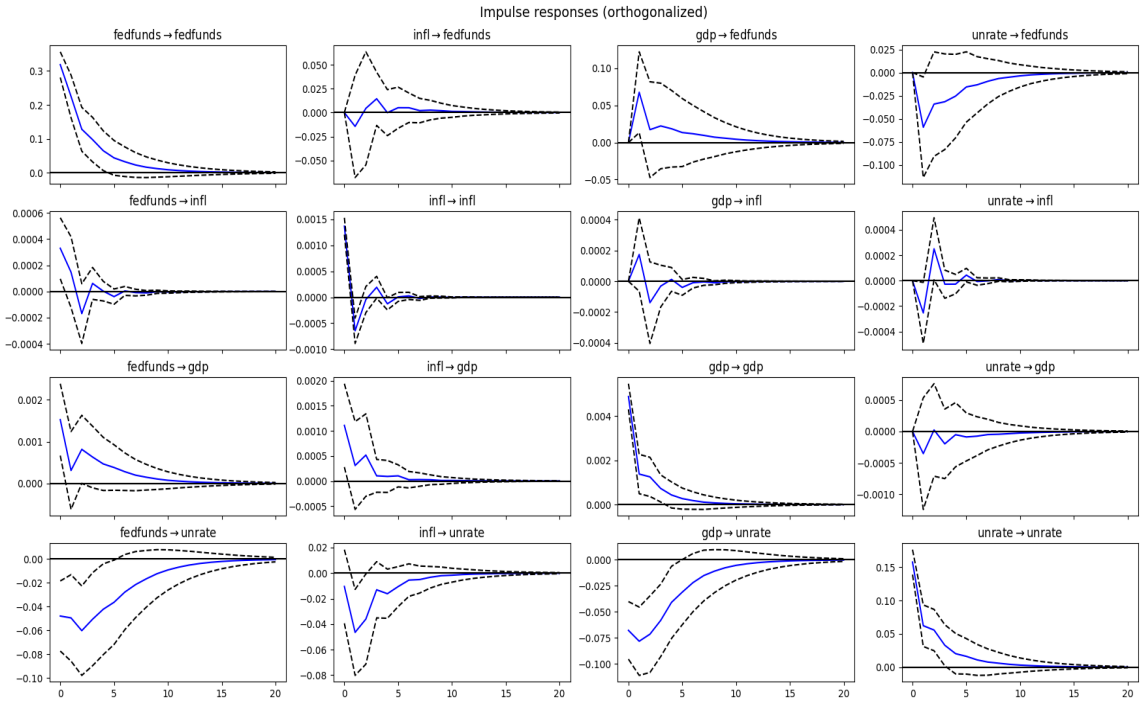


Figure 3: Impulse-Response functions plotted up to 20 quarters.

6 Conclusion

This analysis of the Phillips Curve using a structural VAR model provides insights into the temporal relationship between inflation and unemployment in the U.S. economy. Our findings confirm that, while a negative short-term relationship exists between these variables, it diminishes significantly over time. Specifically, shocks to inflation show a temporary reduction in unemployment, but this effect fades after approximately 10 quarters, with unemployment eventually returning to its natural rate. These results align with previous economic theories suggesting that the trade-off between inflation and unemployment is short-lived, challenging the notion of a long-term exploitable relationship. This study underscores the limitations of monetary policy in sustaining lower unemployment through inflationary strategies, reinforcing the idea that central banks must consider the temporal dynamics of the Phillips Curve when designing policy interventions. Further research could expand on these findings by exploring the impacts of additional macroeconomic variables and examining the robustness of these results in different economic contexts.

7 Bibliography

1. Sims, Christopher A. "Macroeconomics and Reality." Econometrica 48, no. 1 (1980): 1–48.
2. Hamilton, James D. Time Series Analysis. Princeton:Princeton University Press, 1994.
3. McCracken, M.W., Ng, S., 2015; FRED-MD: A Monthly Database for Macroeconomic Research, Federal Reserve Bank of St. Louis Working Paper 2015-012.
4. McCracken, M.W., Ng, S., 2020; FRED-QD: A Quarterly Database for Macroeconomic Research, Federal Reserve Bank of St. Louis Working Paper 2020-005.
5. Romer, David. Advanced Macroeconomics. New York :McGraw-Hill 2012.
6. Dickey, D. & Fuller, Wayne. (1979). Distribution of the Estimators for Autoregressive Time Series With a Unit Root. JASA. Journal of the American Statistical Association. 74. 10.2307/2286348.
7. Box, G. E. P., and David A. Pierce. "Distribution of Residual Autocorrelations in Autoregressive-Integrated Moving Average Time Series Models." Journal of the American Statistical Association 65, no. 332 (1970): 1509–26.
8. Granger, C. W. J. "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods." Econometrica 37, no. 3 (1969): 424–38.
9. Phillips, A.W. (1958), The Relation Between Unemployment and the Rate of Change of Money Wage Rates in the United Kingdom, 1861–19571. Economica, 25: 283-299.
10. Robert E. Lucas, Econometric policy evaluation: A critique, Carnegie-Rochester Conference Series on Public Policy, Volume 1, 1976, Pages 19-46.